

Hongru Liu

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EDUCATION

Carnegie Mellon University, Pittsburgh, PA

Dec 2026

Master of Science in Mechanical Engineering - Advanced Study

Overall GPA: 3.68/4.0

Relevant Coursework: Mechatronic Design, Introduction to Deep Learning, AI Machine Learning, Modern Control Theory, Robot Dynamics and Analysis, Soft Robots: Mechanics, Design and Modeling, Computer Vision for Engineers

Rensselaer Polytechnic Institute, Troy, NY

May 2025

Bachelor of Science in Mechanical Engineering to be conferred May 2025

Overall GPA: 3.70/4.0 | Dean's Honor List (2021-2024)

Relevant Coursework: Inventor's Studio, Fluid Mechanics, Modeling & Control of Dynamic Systems, thermodynamics

RESEARCH EXPERIENCE

Lunar Soil Mix Printing

Sep 2024 - May 2025

Led by Dr. Semih Akin

Rensselaer Polytechnic Institute

- Mixed lunar regolith simulant with resin and fabricated ASTM-compliant tensile bars, enabling strength evaluation across multiple soil ratios (Testing from 20% to 70%), succeeded printed out 20% to 30% sample using clay printer and SLA 3D printer
- Optimized formulations (particle size, resin composition, exposure time) and documented structural performance, providing recommendations to improve printability and material reliability for lunar construction

Thermal Power Dispatch System Design for Nuclear Flexible Plant Operation and Generation

Jan - May 2024

Led by Dr. Shanbin Shi

Rensselaer Polytechnic Institute

- Modeled and simulated thermal dispatch systems in MATLAB Simulink, validating assumptions through thermal-hydraulic literature analysis, accomplished 5% of whole Simulink diagram analysis
- Collaborated with graduate researchers to align methodologies, improving accuracy and efficiency of nuclear flexible plant modeling

PROJECT EXPERIENCE

Mechatronic Design— Hexapod Dancing Robot

Jan 2026 - May 2026

- Iteratively designed a multi-actuator hexapod dancing robot in Fusion 360 from concept to functional prototype, optimizing chassis layout, leg linkage, and component packaging through repeated CAD revisions
- Evaluated multiple chassis, drivetrain, and leg architectures using CAD studies and physical mockups, refining dimensions, interfaces, and structural clearances based on testing feedback
- Performed motor and gearbox trade studies across torque, speed, footprint, and backlash, selecting an integrated drivetrain solution for synchronized leg motion
- Managed prototype sourcing and fabrication, including motors, gearboxes, brackets, fasteners, and wood panels, while coordinating tolerances between purchased and custom-machined parts
- Machined custom structural and electronics mounting panels through drilling, cutting, milling, and wood board fabrication, balancing rigidity, weight, and assembly efficiency
- Supported full mechanical-electrical integration and workshop assembly, including actuator mounting, gearbox alignment, subsystem debugging, and repeated prototype iteration cycles

Modern Control Theory — Autonomous Vehicle & Quadrotor Drones Control

Aug 2025 - Dec 2025

- Designed and implemented model-based control algorithms for autonomous ground vehicles and quadrotor Drones, focusing on trajectory tracking, stability, and robustness
- Developed quadrotor flight controllers using PID, LQR, and adaptive control techniques, incorporating dynamic modeling and state feedback for attitude and position control
- Implemented vehicle lateral and longitudinal controllers for path following and maneuvering, integrating discrete-time control and simulation-based validation
- Evaluated controller performance through time-domain simulations and stability analyses in MATLAB/Simulink and Python

SKILLS

Modeling/Tool: CAD (NX, Fusion, SolidWorks), Excel

Simulation: Finite elements analysis (NX), MATLAB, Simulink

Machining: 3D Printing, Lathe Operations, Milling Machine Operations, Welding, Non-Metallic Fabrication

Language: English (Proficient), Mandarin Chinese (Native)